

Lab Session 6 — Aid Effectiveness

Introduction to Applied Regression Analysis

ICPSR Summer Program 2026

1. Working with World Bank Data

You will be working with data from the World Bank. R has a useful command to directly download data from the World Bank. In order to use this command, you'll first need to download its library (WDI):

```
install.packages("WDI")
library(WDI)

# Download data from the World Bank
aid_data <- WDI(indicator =
  c("DT.ODA.ODAT.GN.ZS", "SE.PRM.CMPT.ZS", "SH.DYN.MORT",
    "SH.STA.MMRT", "SL.EMP.VULN.ZS"),
  country = "all", start = 1960, end = 2020, extra = TRUE)
```

These data examine the effectiveness of aid.

2. You will tidy this dataset, and along the way you will calculate some commonly used transformations of the data.
3. Merge another dataset (`polity4.dta`) to include more variables. Particularly, you are interested in the variable `polity2`. Use the following command:

```
aid_data2 <- merge(aid_data, polity4,
  by = c("iso3c", "year"), all.x = TRUE)
```

4. Identify a continuous variable to proxy aid effectiveness (hints: common proxies include mortality rates, maternal mortality, school enrollment, but others may be fine). Once you find one, check assumptions about normality.
5. Fit a regression with your proxy as the dependent variable, `DT.ODA.ODAT.GN.ZS` (aid received as a percent of GNI) as the main explanatory variable. What are your inferences?
6. Next, include `polity2` as a control. Do your inferences change?
7. Create a new variable, `demdum`, that is a binary measure of democracy. Any country with a `polity2` score of six or higher will be coded as a one. Otherwise a country is not a democracy. Does including this new measure of democracy change your results?
8. Interact democracy and aid.
9. How would you best present these preliminary results in a table and in a graph?

10. Run some of the diagnostics tests discussed earlier. Are there any issues with the models?